Data Science

“Data Science is the most sought-after profession of the 21st century. If data is the new oil, then Data Science is the combustion engine driving the digital revolution.”

Definition

- Data Science is an interdisciplinary field that uses statistics, computer science, and domain expertise to extract insights and knowledge from structured and unstructured data.
- Data science is the science of collecting, storing, processing, describing, and mapping data.

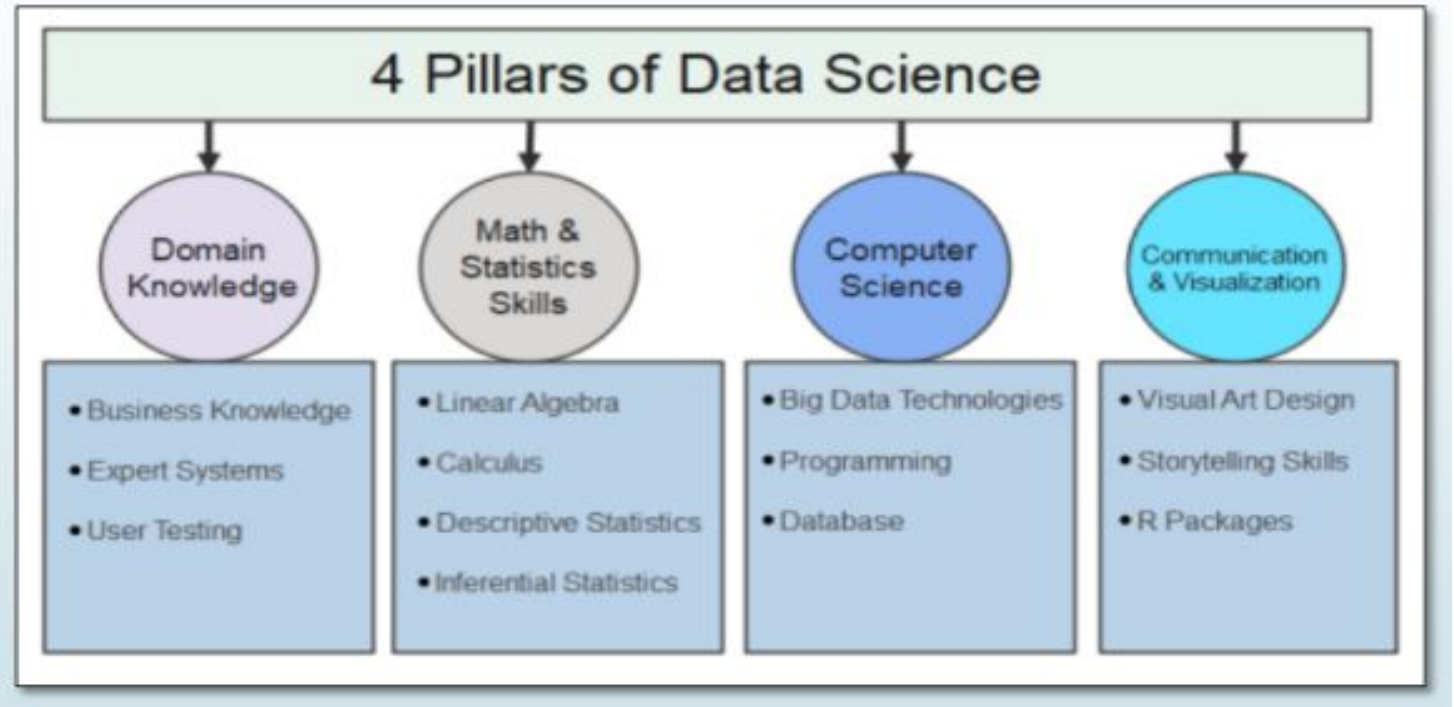
What is Data Scientist?

- A data scientist leads research projects to **extract valuable information from big data** and is skilled in technology, mathematics, business, and communications.
- Organizations use this information to make better decisions, solve complex problems, and improve their operations.
- That's why data scientists are in **high demand** and even considered "rock stars" in the business world.

What does a Data Scientist do?

- A data scientist collects, analyzes, and interprets big data to uncover patterns and insights, make predictions, and create actionable plans.
- Big data can be defined as datasets that have greater variety, volume, and velocity than earlier methods of data management were equipped to handle.

Pillars of Data Science:



- **Computer programming.** Data scientists write queries using languages such as R or Python to pull data from their company's database.
- **Mathematics, statistics, and probability.** Data scientists draw on these skills to analyze data, test hypotheses, and build machine learning models—files that data scientists train to recognize certain types of patterns.
- **Domain knowledge**. To translate data into relevant and meaningful insights that drive business outcomes, data scientists also need domain knowledge—an understanding of the industry and company where they work.
- **Communication & Visualization**. Data science relies on visualization to share ideas and explain data. Improve data comprehension.

Application and results obtained using data science

Application	How Data Science is Used	Results Obtained
Healthcare	Predict diseases, analyze patient records, recommend treatments	Early disease detection, improved patient care, reduced healthcare costs
Finance	Detect fraud, assess credit risk, predict stock trends	Reduced fraud, better investment decisions, faster loan approvals
E-commerce	Recommend products, analyze customer behavior	Higher sales, improved customer satisfaction, personalized shopping
Education	Track student performance, identify learning gaps	Better academic performance, personalized learning
Transportation	Optimize routes, predict traffic, manage logistics	Reduced travel time, lower fuel costs, efficient delivery

Overall Results Obtained Using Data Science

- Improves decision-making through data-driven insights.
- Increases efficiency and productivity.
- Reduces costs and saves time.
- Detects fraud and security threats.
- Enhances customer satisfaction through personalization.
- Predicts future trends and outcomes.
- Automates repetitive tasks.
- Supports innovation and business growth.

Tools and Technologies

Languages

- **Python, R, SQL**

Libraries

- pandas, NumPy, scikit-learn **PostgreSQL**
- TensorFlow, PyTorch

Tools

Jupyter, Tableau, Power BI

Apache Spark, Hadoop

Databases

MySQL, MongoDB,

Overview of the Data Science Process

- Data Science Methodology indicates the routine for **finding solutions to a specific problem.** This is a cyclic process that undergoes a critic behaviour guiding business analysts and data scientists to act accordingly.
- Data Science methodology is a structured approach to solving complex problems using data.
- The following are the typical steps involved in a Data

1. **Business Understanding:**
- The business **problem** is defined,
- The **objective** of the analysis is identified.
- The data science team should **work closely** with the business stakeholders to understand the problem and define the goals.
2. **Data Understanding:**
- The data science team identifies and collects the data required for the analysis.
- They explore the data to understand its structure, quality, and completeness.
3. **Data Preparation:**
- In this stage, the data science team cleans, transforms, and prepares the data for analysis.
- This step is critical to ensure that the data is in the right format and quality for analysis.

4.Data Modelling:

- In this stage, the data science team selects the appropriate modelling techniques to analyze the data and build predictive models.
- This stage involves selecting the right algorithms, tuning the model parameters, and validating the model.

5. Evaluation:

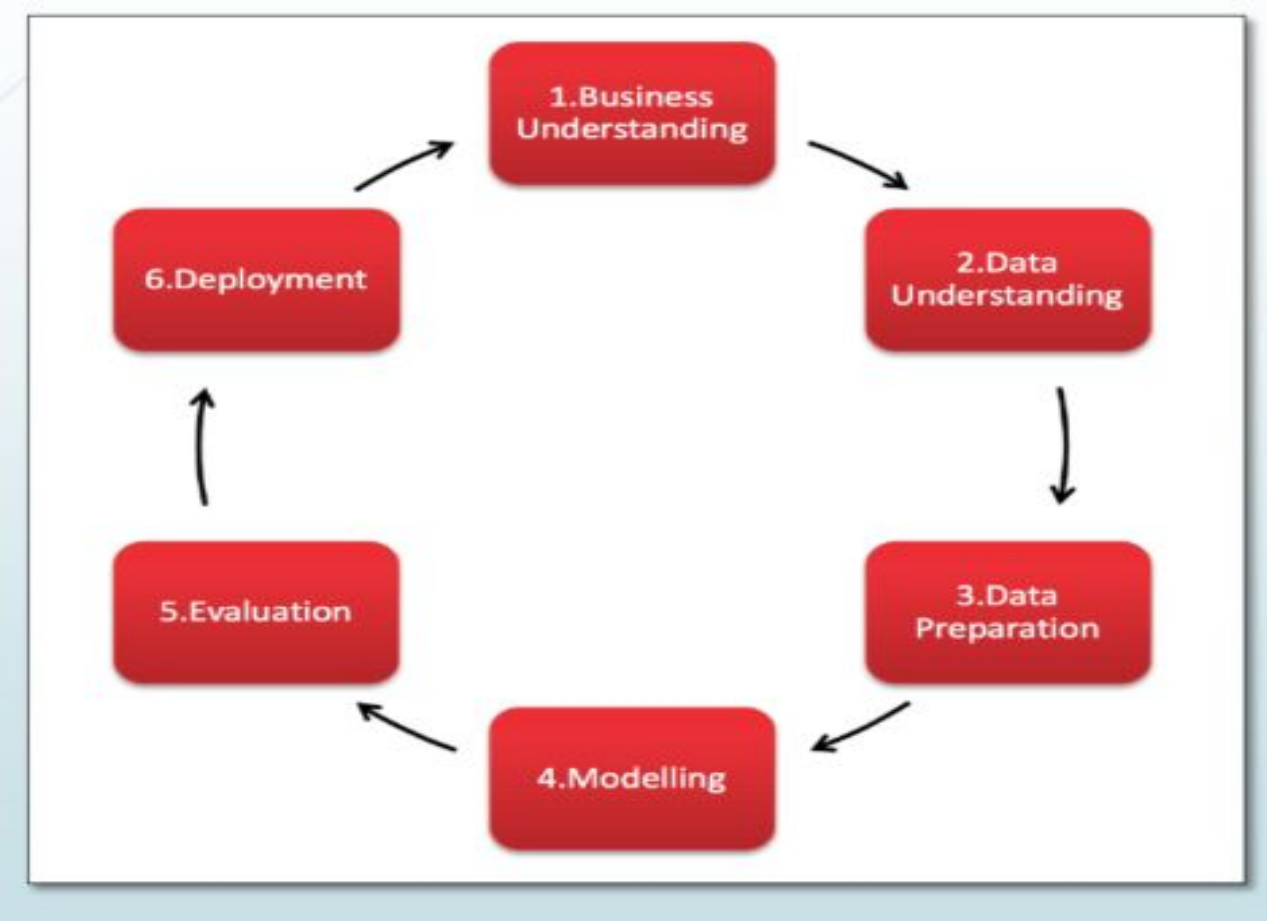
- In this stage, the data science team evaluates the model's performance and its ability to solve the business problem.
- They use various evaluation metrics to determine the effectiveness of the model and make improvements if necessary.

6. Deployment:

In this stage, the data science team deploys the model in the production environment, integrating it into the business processes, and ensuring that it is working correctly.

7. Monitoring and Maintenance:

In this stage, the data science team monitors the model's performance in the production environment, making necessary changes and improvements to ensure that it continues to work effectively.


1.Business Understanding
2.Data Understanding
3.Data Preparation
4.Modelling
5.Evaluation
6.Deployment

Types of Data

- Data is the raw form of information, a collection of facts, figures, symbols or observations that represent details about events, objects or phenomena.
- By itself, data may appear meaningless, but when organized, processed and interpreted, it transforms into valuable insights that support decision- making, problem-solving and innovation.

Data categories into following types

- Quantitative data
- Qualitative data
- Structured data
- Unstructured data
- Semi-structured data

1. Quantitative Data

- Quantitative data is information that can be measured, counted and expressed in numerical form. It provides **objective values** that can be analyzed statistically to **identify patterns, trends and relationships.**
- Represents numbers and measurable values.
- Can be divided into: Discrete data (Whole numbers) and Continuous data (Values on a scale).
- Widely used in research, finance, engineering and business analytics.
- Example: Age of people, number of customers visiting a store, temperature readings, sales revenue.

2. Qualitative Data

- Qualitative data is descriptive, non-numeric information that explains qualities, characteristics or categories rather than quantities. It helps understand opinions, experiences and meanings behind behaviors.
- Focuses on qualities, attributes and categories rather than numbers.
- Often collected **through surveys, interviews or observations.**
- Useful for understanding opinions, motivations and behaviors.
- Example: Customer feedback (“satisfied”, “unsatisfied”), product colors, interview
- transcripts, social media comments.

3. Structured Data

- Structured data is information organized into a predefined format (rows and columns) that makes it easily searchable and manageable by traditional databases.
- Stored in relational databases or spreadsheets.
- Easy to process with SQL and other tools.
- Best suited for tasks requiring accuracy and consistency.
- Example: Bank transactions, employee records, product inventories.

4. Unstructured Data

- Unstructured data is raw information that does not follow a predefined structure or format making it harder to organize and analyze with conventional tools.
- Accounts for over 80% of data generated globally.
- Requires advanced tools (AI, NLP, computer vision) to extract insights.
- Common in social media, multimedia and IoT applications.
- Example: Emails, images, videos, voice recordings, PDF documents.

5. Semi-Structured Data

- Semi-structured data combines aspects of structured and unstructured data. It does not reside in traditional tables but still contains tags or markers that provide a loose structure.
- Provides a balance between flexibility and structure.
- Easier to analyze than unstructured data, but less rigid than structured data.
- Often used in web applications, IoT devices and log systems.
- Example: JSON files, XML documents, NoSQL databases, sensor logs.

Representation of Data in Data Science

1. Tables

- Data is arranged in rows and columns.
- Used for storing and organizing structured data.

2. Charts and Graphs

- **Bar Chart:** Compares values across different categories.
- **Line Graph:** Shows trends or changes over time.
- **Pie Chart:** Displays proportions or percentages of a whole.
- **Scatter Plot:** Shows relationships between two numerical variables.

3. Images

Data can also be represented as images, such as:

- Medical scans (X-rays, MRI)
- Satellite images
- Photographs used in computer vision

4. Text

Unstructured data represented as:

- Emails
- Reviews
- Social media posts
- News articles

Text data is analyzed using **Natural Language Processing (NLP)**.

5. Time Series

Data collected over time at regular intervals.

Examples:

- Daily stock prices
- Monthly sales
- Hourly weather readings

Importance of Data Representation

- Makes data easier to understand.
- Helps identify patterns and trends.
- Supports better decision-making.
- Improves communication of insights through visualizations.
- Enables effective analysis and machine learning.

Data Science Process Life cycle

Data Collection

A Key Step in the Data Science Lifecycle

Air Pollution Prediction in Delhi

Data Sources and Methods

- CPCB (Central Pollution Control Board) API – Real-time air pollutant data (PM2.5, PM10, NO2, CO) via REST API.
- IoT Sensors – Temperature, humidity, and wind speed from rooftop devices (via MQTT/HTTP).
- Google Maps API – Live traffic congestion data to estimate vehicular emissions.
- Twitter API – Public complaints and smog alerts using hashtag-based tweet streaming.
- NASA Satellite Feeds – Regional aerosol density and weather info from satellite imagery.

Usage

- The collected data is fed into machine learning models for short-term pollution forecasting and public health advisories.

Real-Time Data Collection: OpenAQ API

Steps to Use OpenAQ API

1 Register: Sign up at https://accounts.openaq.org and create a free account.

2 Get API Key: Go to the dashboard → API Keys → Generate a key.

3 Set Headers: Include the key in request headers using X-API-Key.

4 Make Request: Use endpoints like /v3/measurements to fetch data.

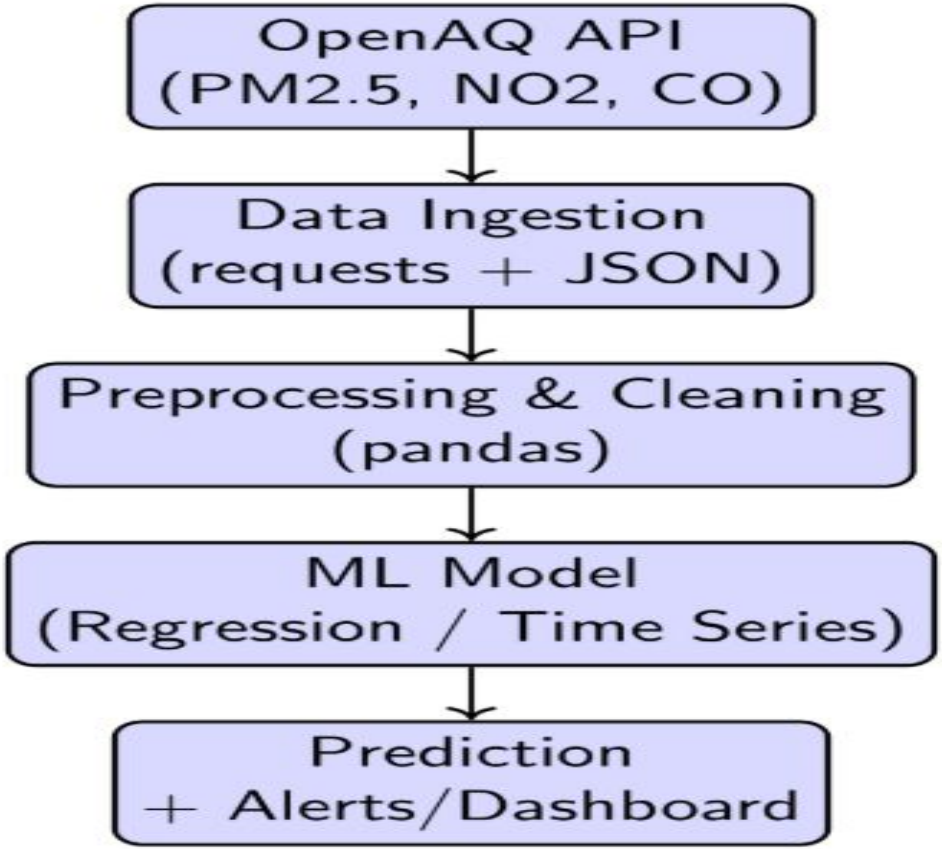
Example: Python Script to Fetch PM2.5 Data (Delhi)

Python Code

```
import requests
url = "https://api.openaq.org/v3/measurements"
params = {
        "country": "IN",
        "city": "Delhi",
        "parameter": "pm25",
        "limit": 5,
        "sort": "desc"
}
headers = {
        "accept": "application/json",
        "X-API-Key": "YOUR_API_KEY" # Replace with your actual key
}
```

```
response = requests.get(url, params=params, headers=headers)
if response.status_code == 200:
    data = response.json()
    for result in data["results"]:
    print(f"{result['location']} | PM2.5: {result['value']} {result['unit']} | Time:
{result['date']['utc']}")
Else:
    print("Error:", response.status_code, response.text)
```

Real-Time Air Quality Monitoring: Data Pipeline



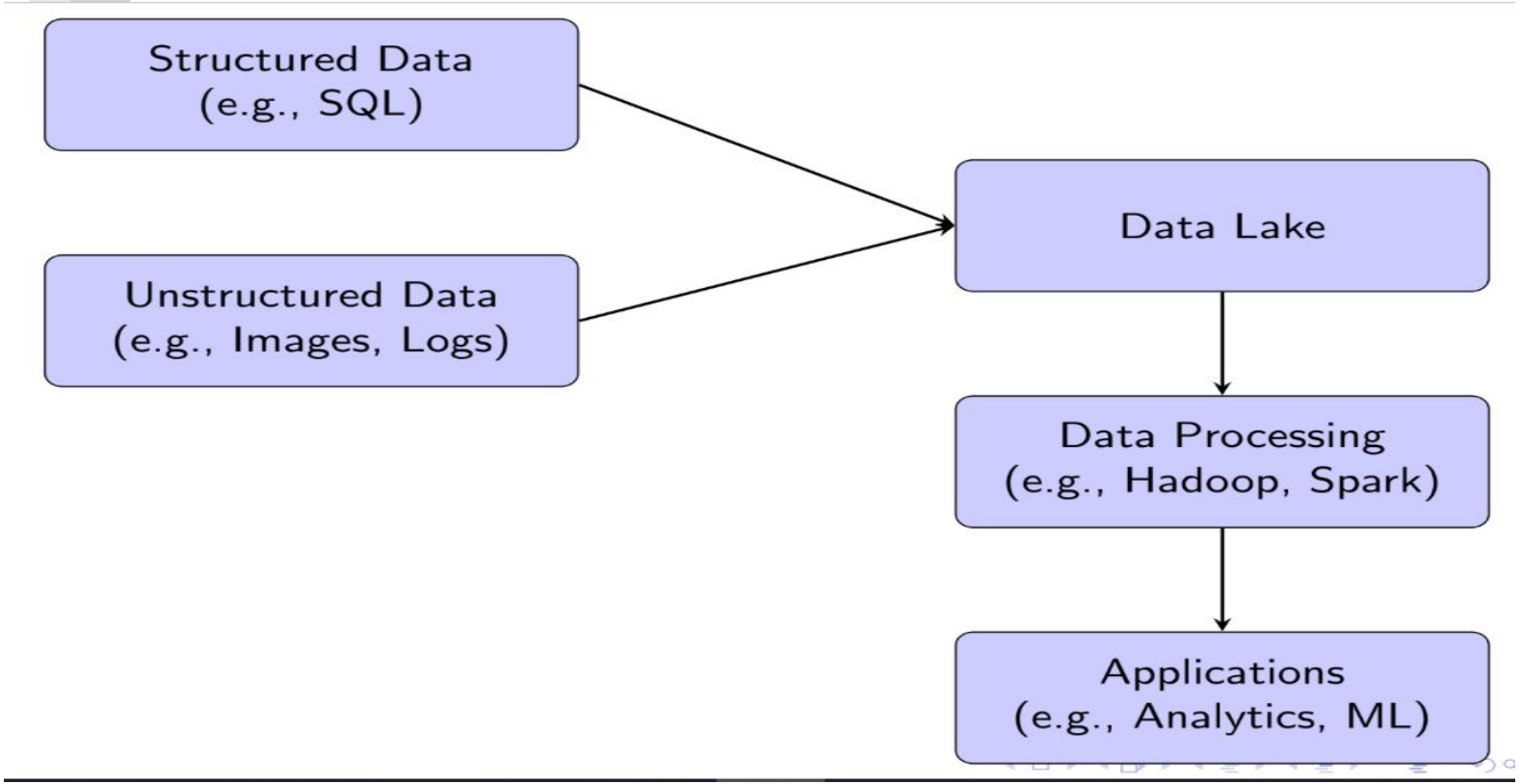
Data Storage

A Key Step in the Data Science Lifecycle

Types of Data Storage in Data Science

- Structured Data: Organized into rows and columns, stored in relational databases (e.g., MySQL, PostgreSQL).
- Unstructured Data: No fixed format; includes text, images, videos, etc. (e.g., social media posts, emails).
- Data Lake: Centralized repository storing structured and unstructured data at any scale (e.g., Hadoop, Amazon S3).

Application Sketch: Data Storage Workflow



Real-Time Application Examples

Data Type	Format	Storage	Application
Structured	Tabular	MySQL, Oracle	Fraud Detection, Statement Generation, Banking Transactions
Unstructured	Text, Images, Videos	MongoDB, Amazon S3	Sentiment Analysis, Trend Prediction
Data Lake	Structured + Unstructured	Amazon S3, Hadoop	Recommendation Systems, E-commerce Platform

Processing Data

A Key Step in the Data Science Lifecycle

Process and Parse Data includes:

Data Cleaning

- Remove duplicates
- Handle missing values
- Correct errors

```
import pandas as pd

# Load dataset
data = pd.read_csv("students.csv")

# Remove duplicate rows
data = data.drop_duplicates()

# Fill missing values
data = data.fillna(0)

# Save cleaned data
data.to_csv("cleaned_students.csv", index=False)

print(data)
```

Data Parsing

- Data parsing is the process of analyzing raw, unstructured data and converting it into a clean, structured format like JSON, XML, or CSV so software can read it.
- Raw data is transformed into a clean and organized format for analysis.
- **Steps:**
  a. Clean the data.
  b. Convert data types (text to numbers, dates, etc.).
  c. Remove unnecessary information.
  d. Organize data into tables or databases.

Example 1: Parsing a JSON File

Raw JSON Data (student.json)

```
{
  "name": "Rahul",
  "age": 21,
  "course": "Data Science",
  "marks": 88
}
```

Python Code

```
import json

# Open JSON file
with open("student.json", "r") as file:
    student = json.load(file)

# Access parsed data
print("Name:", student["name"])
print("Age:", student["age"])
print("Course:", student["course"])
print("Marks:", student["marks"])
```

Output

Name: Rahul

Age: 21

Course: Data Science

Marks: 88

Explanation:

The JSON parser reads the raw JSON file and converts it into a Python dictionary so that each value can be accessed easily.

Data Manipulation

Data manipulation in data science is the process of **modifying, organizing, transforming, and restructuring data** to make it suitable for analysis, visualization, and machine learning.

Common Data Manipulation Tasks

- **Filtering** – Select only the required rows or columns.
- **Sorting** – Arrange data in ascending or descending order.
- **Grouping and Aggregation** – Group data by categories and calculate statistics such as sum, average, count, minimum, or maximum.
- **Merging and Joining** – Combine data from multiple tables or sources.

Data Manipulation

1. Filtering

Definition: Filtering is used to **select only the records that meet a specific condition**.

- **Example:** Show students with marks greater than 80.

```
import pandas as pd
data = pd.DataFrame({
    "Name": ["Rahul", "Priya", "Amit"],
    "Marks": [85, 92, 75]
})
filtered = data[data["Marks"] > 80]
print(filtered)
```

Sorting

- Sorting arranges data in **ascending or descending order**.
- **Example:** Sort students by marks (highest to lowest).

sorted_data = data.sort_values(by="Marks", ascending=False) print(sorted_data)

Merging

- Merging combines two or more datasets using a common column.

```
students = pd.DataFrame({
    "ID": [1, 2],
    "Name": ["Rahul", "Priya"]
})
marks = pd.DataFrame({
    "ID": [1, 2],
    "Marks": [85, 92]
})
result = pd.merge(students, marks, on="ID")
print(result)
```

Aggregation

- Aggregation summarizes data using functions like **sum, average, count, maximum, and minimum**.

```
salary = pd.DataFrame({
    "Department": ["HR", "HR", "IT", "IT"],
    "Salary": [30000, 35000, 50000, 55000]
})
avg_salary = salary.groupby("Department")["Salary"].mean()
print(avg_salary)
```

Data Transformation

- Data transformation changes data into a more useful or meaningful format.

```
data["Grade"] = data["Marks"].apply(
    lambda x: "A" if x >= 80 else "B"
)
print(data)
```

Data Cleaning

- **Data Cleaning** is the process of identifying and correcting errors in a dataset to improve its quality before analysis. Clean data produces more accurate and reliable results.

1. Handling Missing Values :

   Missing values are data fields that are empty or contain no value. They can be removed or replaced.

```
import pandas as pd

data = pd.DataFrame({

    "Name": ["Rahul", "Priya", "Amit"],

    "Marks": [85, None, 78]

})

# Fill missing values with average

data["Marks"] = data["Marks"].fillna(data["Marks"].mean())

print(data)
```

Removing Duplicates

- Duplicate records are repeated entries in a dataset. Removing them improves data accuracy.

```
data = data.drop_duplicates()
print(data)
```

Correcting Inconsistent Data

- Inconsistent data occurs when the same information is written in different formats. Standardizing the format makes the data consistent.

```
data["City"] = data["City"].str.title()
print(data)
```

Detecting and Treating Outliers

- Outliers are values that are **much higher or lower** than the rest of the data and may affect analysis.

```
# Remove values greater than 100
clean_data = data[data["Marks"] <= 100]
print(clean_data)
```

Exploratory data analysis

Definition:

Exploratory Data Analysis (EDA) is the process of **examining, summarizing, and visualizing a dataset** to understand its characteristics, discover patterns, detect anomalies (outliers), identify relationships between variables, and prepare the data for further analysis or machine learning.

Objectives of EDA

- Understand the structure and quality of the data.
- Identify missing values, duplicate records, and outliers.
- Discover patterns, trends, and relationships among variables.
- Test assumptions before building machine learning models.
- Select relevant features for analysis.

Steps in Exploratory Data Analysis

Understand the Dataset

- Check the number of rows and columns.
- Identify data types (numeric, categorical, date, etc.).
- Understand the meaning of each variable.

Data Cleaning

- Handle missing values.
- Remove duplicate records.
- Correct inconsistencies and errors.
  -

Summary Statistics

- Calculate measures such as:
  - Mean
  - Median
  - Mode
  - Minimum and Maximum
  - Standard Deviation
  - Variance

Univariate Analysis

- Analyze one variable at a time.
- Use histograms, bar charts, box plots, and frequency tables.

Bivariate and Multivariate Analysis

- Study relationships between two or more variables.
- Use scatter plots, correlation matrices, and grouped charts.

Detect Outliers

- Identify unusual or extreme values using box plots or statistical methods.

Data Visualization

- Create charts and graphs to better understand the data.
- Common visualizations include:
  - Histogram
  - Bar Chart
  - Pie Chart
  - Line Graph
  - Scatter Plot
  - Box Plot
  - Heatmap

Example: COVID-19 Cases per Country

```
countries = ['USA', 'India', 'Brazil', 'UK']
cases = [32000, 28000, 21000, 15000]
plt.bar(countries, cases, color='skyblue')
plt.title('COVID-19 Cases per Country')
plt.xlabel('Country')
plt.ylabel('Number of Cases')
plt.show()
```

ii) Summarization of Data

Definition:

Statistical summarization helps condense data into key figures like mean, median, and standard deviation.

```
scores = [88, 76, 92, 85, 69, 94, 78]
mean_score = np.mean(scores)
median_score = np.median(scores)
std_dev = np.std(scores)
print(f"Mean:  mean_score, Median:  median_score, Std_Dev:  std_dev")
```

Zomato Restaurants Data

- Sourced from Zomato API and Open dataset 1
- Format: CSV, suitable for data cleaning, EDA, and modeling.
- Contains attributes such as:
  - ❖ Name, Location, Country, City
  - ❖ Cuisines, Rating, Votes, Cost for Two
  - ❖ Delivery and Booking Flags
- **Data Cleaning**: Handled missing values, removed duplicates.
- **EDA**: Used Pandas, Matplotlib, Seaborn.
- **Feature Engineering**: Encoded categorical data, grouped cuisines.
- **Modeling:** Optional - clustering, classification, or recommendation.

Applications of Data Science

- Fraud Detection in Banking
- Predictive Maintenance in Industry
- Personalized Recommendations
- Healthcare Diagnosis and Drug Discovery
- Image and Speech Recognition
- Climate Modeling and Forecasting

Related Fields

- Machine Learning – Algorithms and prediction
- Big Data – Handling massive datasets
- Artificial Intelligence – Decision making, planning, reasoning
- Business Analytics – Data-driven business decision making